

*"Write here, write now": Spelling difficulty
disrupts parallel planning in sentence production*

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Slides: github.com/jensroes/slides

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To what extent does difficulty with word spelling affect our ability to convey ideas in writing? This research aims to ...

- ▶ establish when during sentence production word features are being retrieved.
- ▶ evaluate how spelling difficulty affects our ability to plan language.

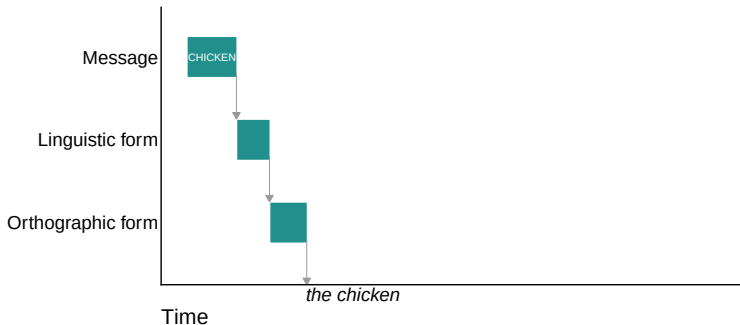
How are levels of language planning coordinated in real time?

Planning language takes time!



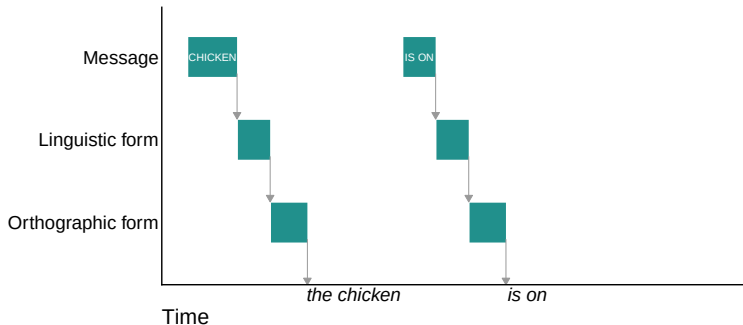
Serial view: Speaker / writers pause at linguistic edges to plan upcoming information.

Planning language takes time!



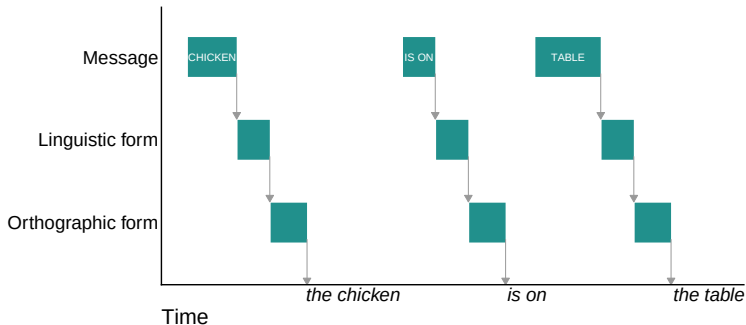
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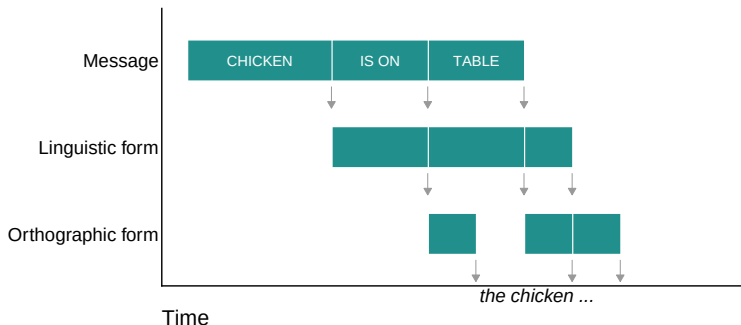
Planning language takes time!



Parallel view: Planning unfolds in parallel to output (Olive, [2014](#); Roeser et al., [2025](#); Van Galen, [1991](#)).

Pauses during production result from delays in information flow! Slow orthographic retrieval delays upstream processes.

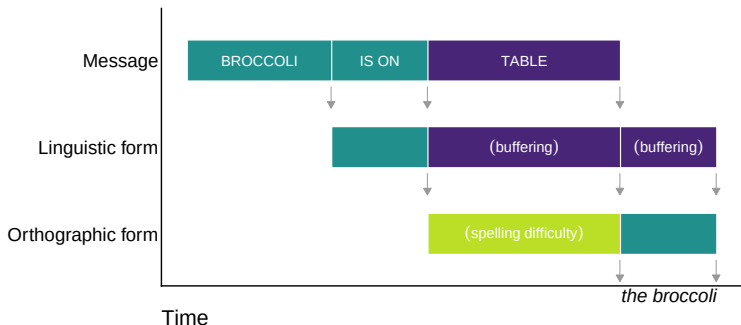
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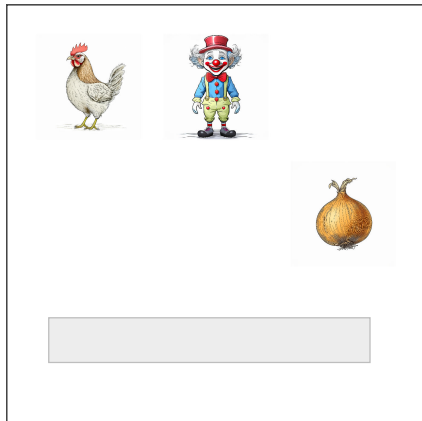
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Parallel view: Planning unfolds in parallel to output (Olive, 2014; Roeser et al., 2025; Van Galen, 1991).

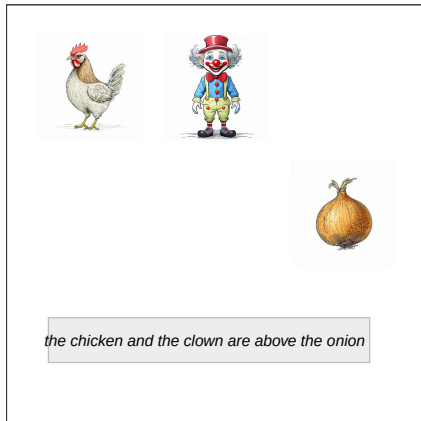
Pauses during production result from delays in information flow! Slow orthographic retrieval delays upstream processes.

Describe the arrangement from left to right!



Example screen (Roeser et al., [2019](#); Smith & Wheeldon, [1999](#))

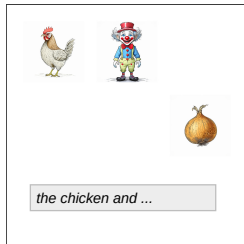
Describe the arrangement from left to right!



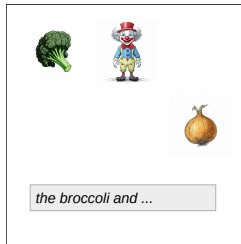
Example screen (Roeser et al., [2019](#); Smith & Wheeldon, [1999](#))

Design: Example item set

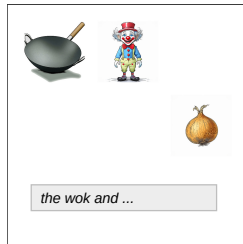
N1: easy spelling / long



N1 difficult spelling / long



N1: easy spelling / short



Prediction: Typing hesitations for N2 (*clown*) result from (1) when N1 spelling (*broccoli*) couldn't be retrieved in time and (2) when N1 length (*wok*) didn't allow for parallel planning.

- ▶ 36 items counterbalanced for prime structure (simple, conjoined subject) and N1 word type.
- ▶ 24 fillers with structures different from items.
- ▶ N1 word type manipulation: easy spelling (long), easy spelling (short), difficult spelling (long):
 - ▶ Long: ≥ 2 syllables
 - ▶ Difficult: spelling diversity $H > 0.5$

What is “spelling diversity”?

Write the name of what you see!

What is “spelling diversity”?



What is “spelling diversity”?

Example summary for “raccoon”(N = 97 participants).

response	N_{spell}	N_{name}	Pr_{spell}	H_{spell}
raccoon	41	75	.55	1.31
racoon	30	75	.40	1.31
raccon	2	75	.03	1.31
raccoo	1	75	.01	1.31
racoo	1	75	.01	1.31

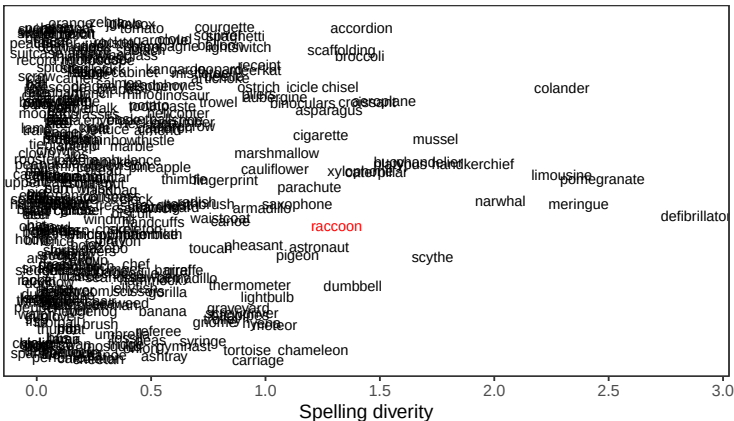
Note. H = diversity statistic; Pr = proportion of responses.

Spelling diversity H_{spell} represents the frequency of a spelling used relative to the number different spellings (Lachman, 1973; Torrance et al., 2018).

$$H_{spell} = \sum_{i=1}^K p_i \times \log_2\left(\frac{1}{p_i}\right), \quad (1)$$

where k is the number of spellings for a picture name; p_i is the proportion of participants producing the i^{th} spelling.

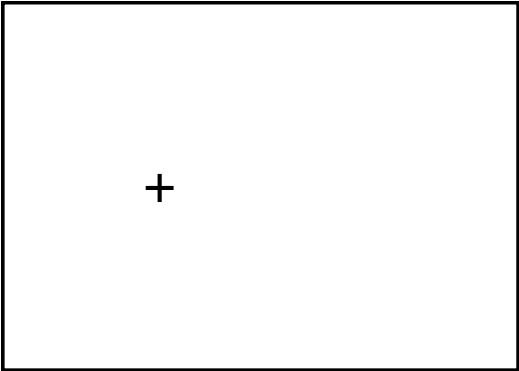
What is “spelling diversity”?



Ten ppts per counterbalancing list. After data cleaning:

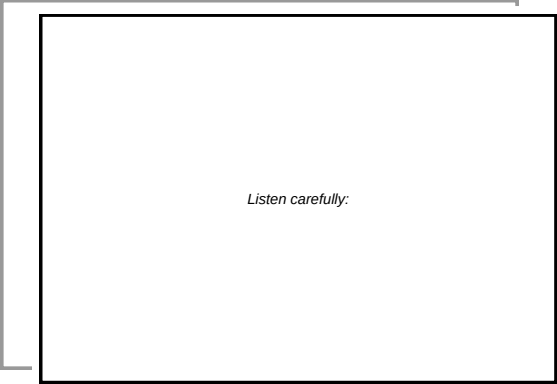
- ▶ Experiment 1: 56 ppts (6 lists)
- ▶ Experiment 2: 80 ppts (9 lists)
- ▶ Experiment 3: 96 ppts (12 lists)

Trial sequence



+

Trial sequence



Listen carefully:

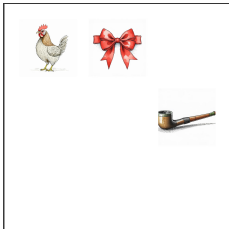
Trial sequence

"The chicken and the bow are above the pipe."

Listen carefully:

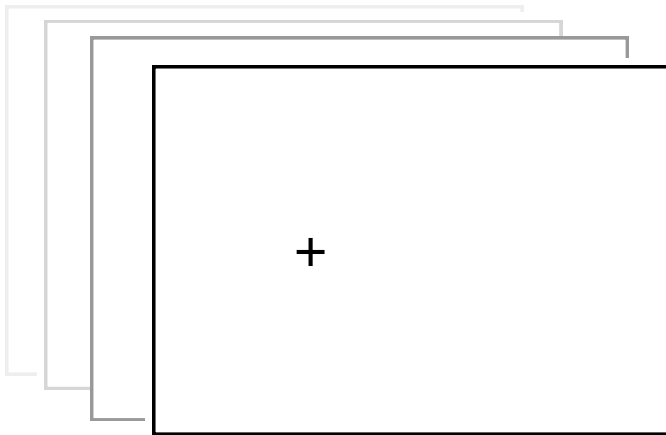
Trial sequence

Does the image on the screen match what you heard?

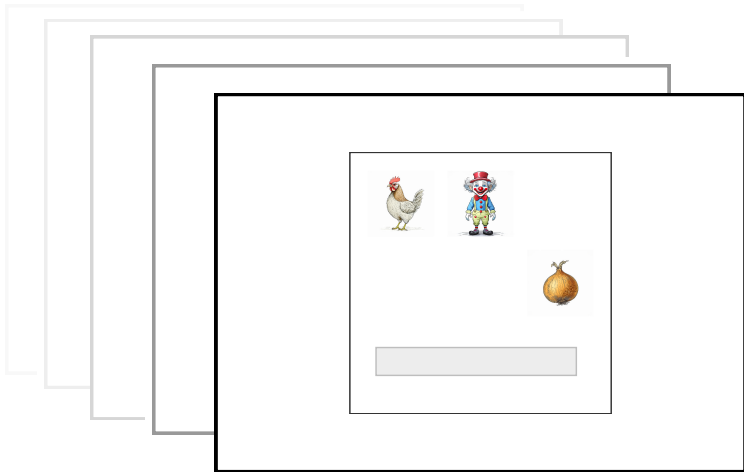


Press Y (Yes) or N (No)

Trial sequence



Trial sequence

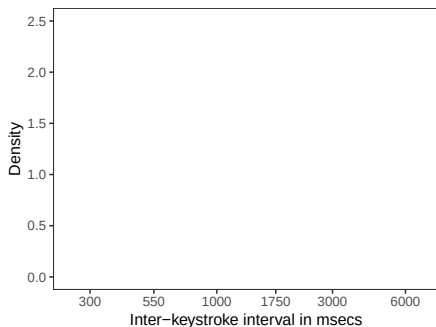


Trial sequence

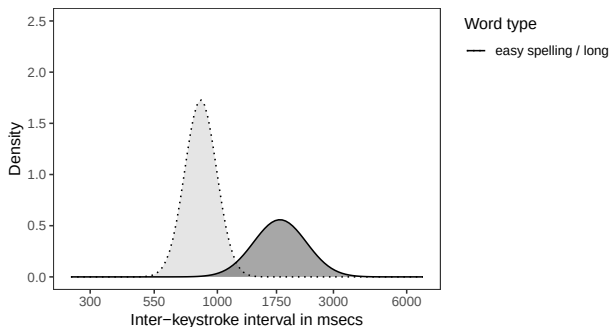


the chicken and ...

Data analysis: typing as mixture process

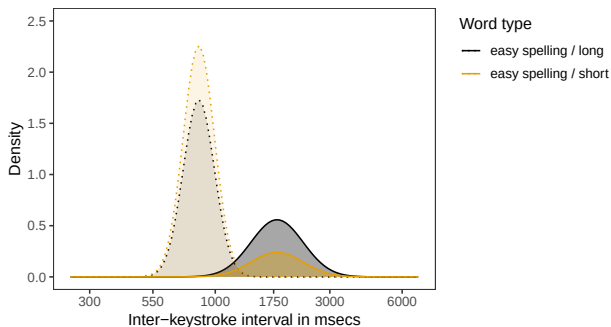


Inter-keystroke intervals were modelled as Bayesian mixed-effects mixture model (Roeser et al., 2024, 2025). For a tutorial see: <https://rpubs.com/jensroes/mixture-models-tutorial>.



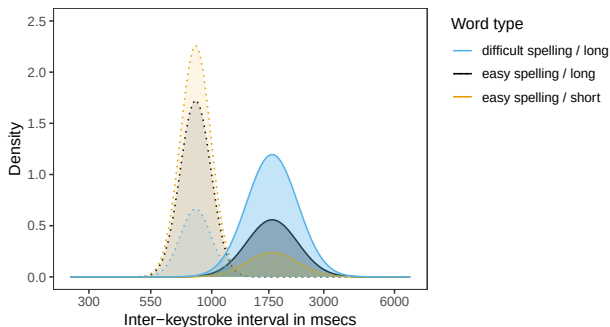
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Data analysis: typing as mixture process



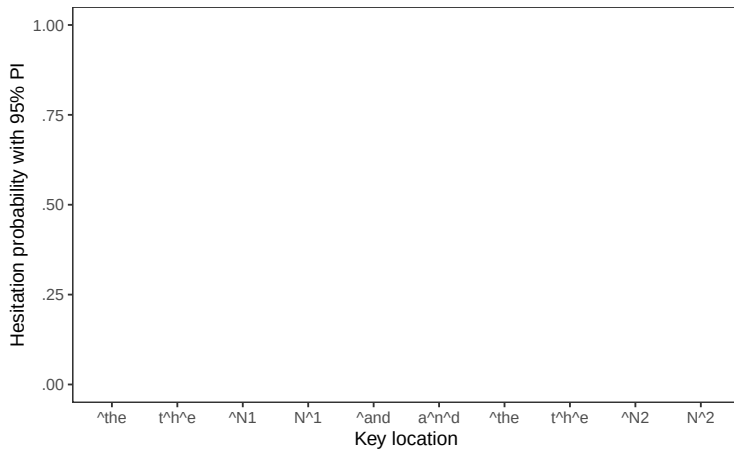
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Data analysis: typing as mixture process

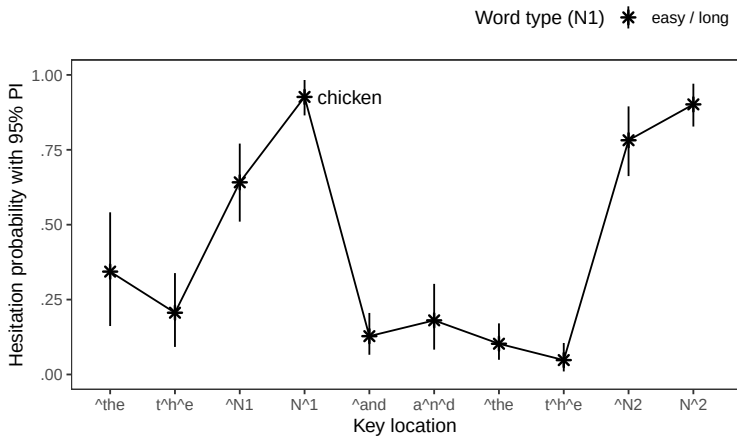


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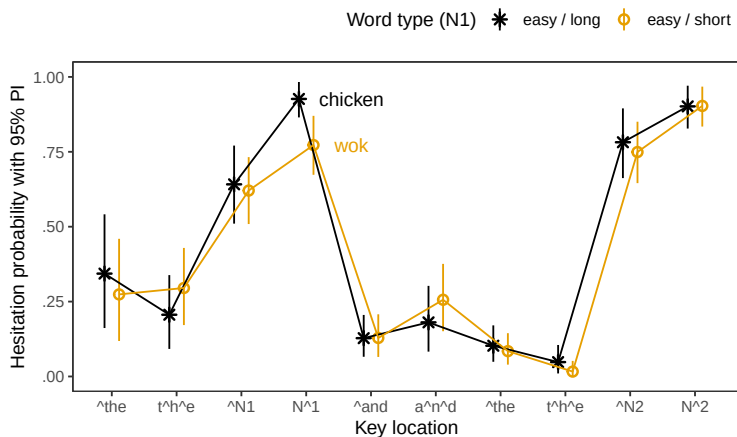
Experiment 1: typing timecourse



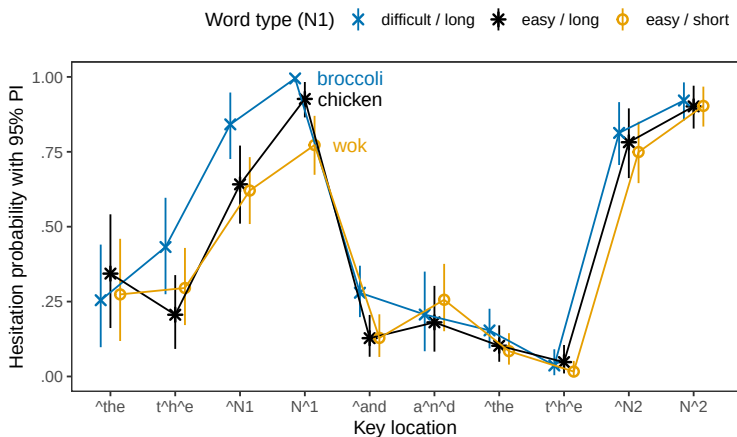
Experiment 1: typing timecourse



Experiment 1: typing timecourse



Experiment 1: typing timecourse



Experiment 2

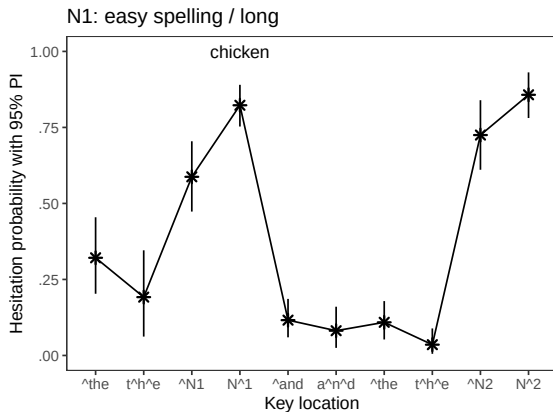
Does typing and the buy writers enough time to avoid hesitations before N2?

Additional prime condition; two target structures:

- ▶ With N2 determiner: *The N1 and the N2 are above the N3*
- ▶ Without N2 determiner: *The N1 and N2 are above the N3*

Prediction: If the distance between nouns constrains parallel planning, we expect more hesitations before N2 when *the* was omitted.

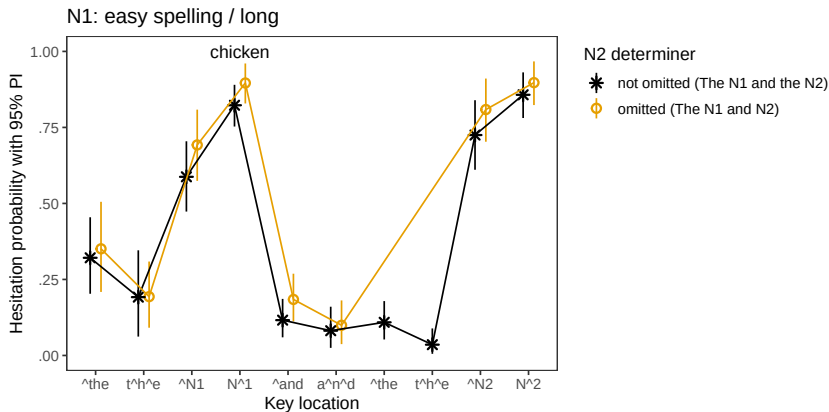
Experiment 2: typing timecourse



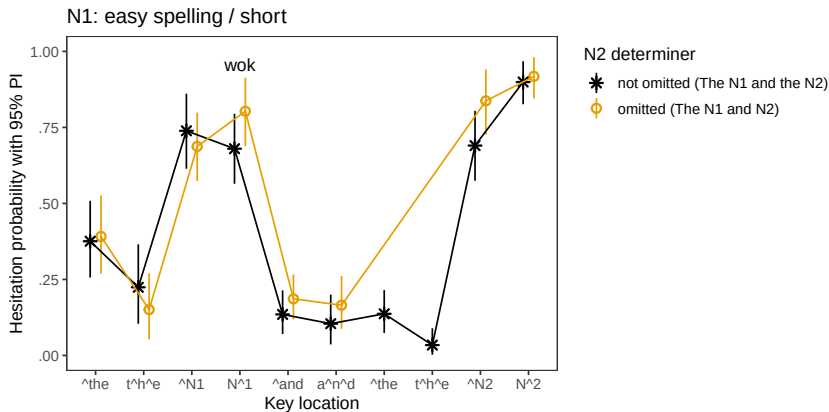
N2 determiner

* not omitted (The N1 and the N2)

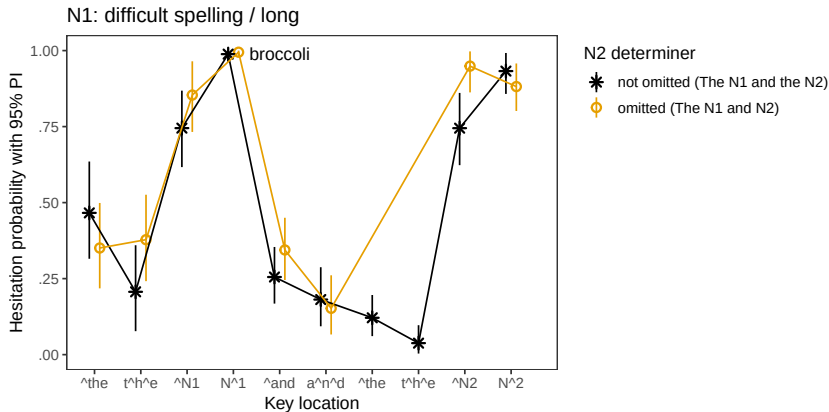
Experiment 2: typing timecourse



Experiment 2: typing timecourse



Experiment 2: typing timecourse



Experiment 3

Are hesitations related to delayed N2 retrieval?

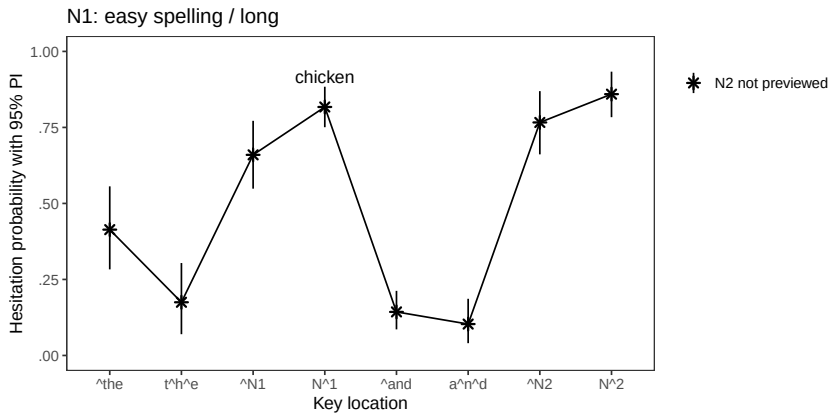
Prime with (auditory) N2 preview:

- ▶ Previewed: *The N1 and clown are above the pipe*
- ▶ Not previewed: *The N1 and bow are above the pipe*

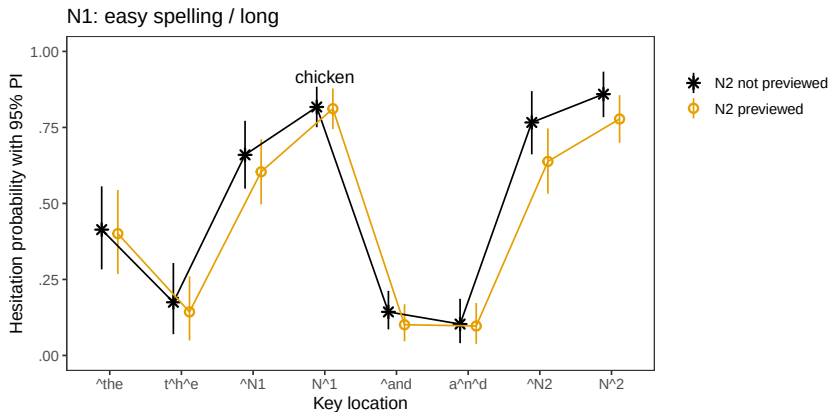
Target: *The N1 and clown are above the onion*

Prediction: If the planning bottleneck delayed N2 retrieval, preview will reduce associated hesitations.

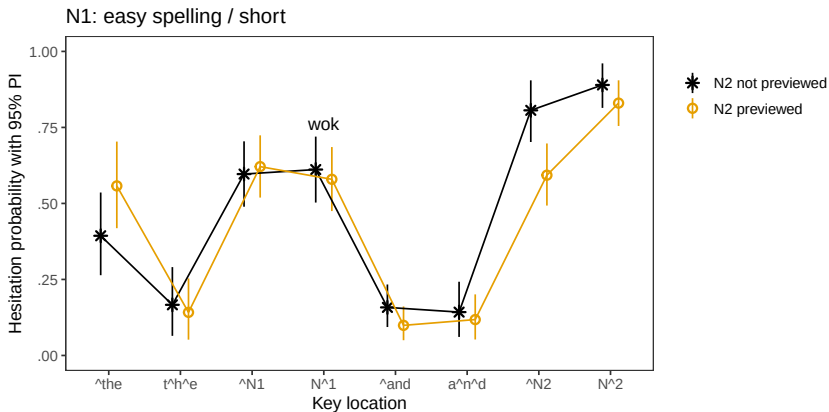
Experiment 3: typing timecourse



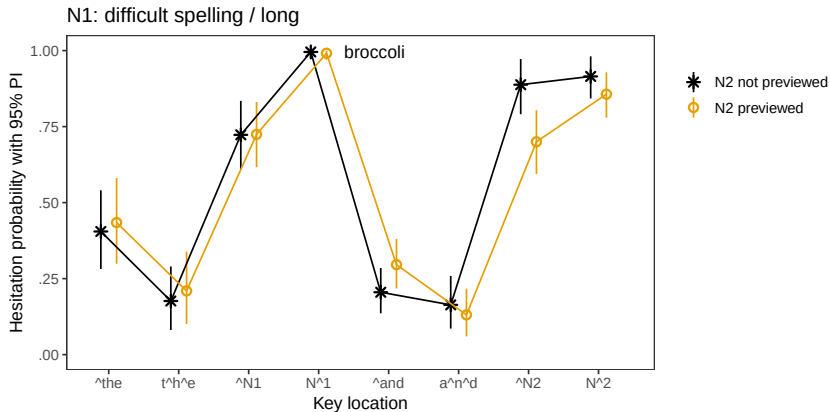
Experiment 3: typing timecourse



Experiment 3: typing timecourse



Experiment 3: typing timecourse



- ▶ Hesitations related to N1 spelling affect N2 retrieval when there was not enough time between N1 and n2 for parallel lexical retrieval of N2.
- ▶ Long and / or functional words – e.g. *and the* – provide time for parallel lexical retrieval (see also Griffin, [2003](#)).
- ▶ Written sentence planning unfolds in parallel to production: difficulty with spelling creates a planning bottleneck with knock-on effects on higher level planning.
- ▶ Serial models of language planning cannot explain these results (Flower & Hayes, [1980](#); Hayes, [2012](#)).

Thanks for listening!

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slides are here: <https://github.com/jensroes/slides>

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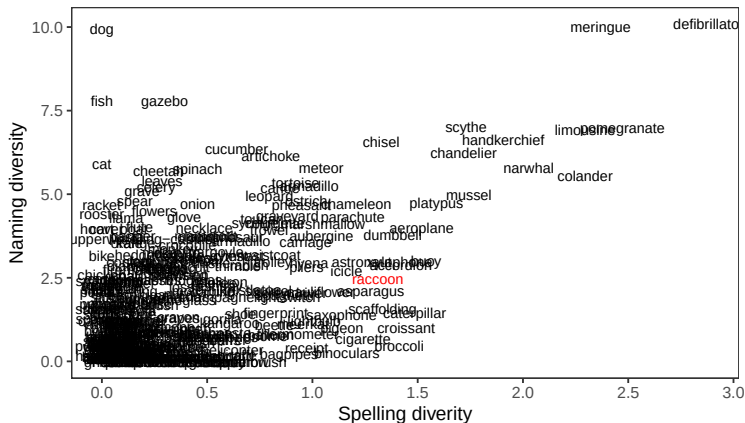
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- Torrance, M., Nottbusch, G., Alves, R. A., Arfé, B., Chanquoy, L., Chukharev-Hudilainen, E., Dimakos, I., Fidalgo, R., Hyönä, J., Jóhannesson, Ó. I., et al. (2018). Timed written picture naming in 14 European languages. *Behavior Research Methods*, 50, 744–758.

References III

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Spelling and naming diversity



N of pics: 383

Typing as mixture process

Inter-keystroke intervals were analysed in mixture models with shifted-lognormal distributions (Roeser et al., [2024](#), [2025](#)).

$$(\log(iki_{ij}) - shift_i) \sim \theta_{condition} \times \mathcal{N}(\beta_{location} + \delta_{location} + u_i + w_j, \sigma_{e'}^2) + \\ (1 - \theta_{condition}) \times \mathcal{N}(\beta_{location} + u_i + w_j, \sigma_e^2)$$

where:

$$u_i \sim \mathcal{N}(0, \sigma_u^2)$$

$$w_j \sim \mathcal{N}(0, \sigma_w^2)$$

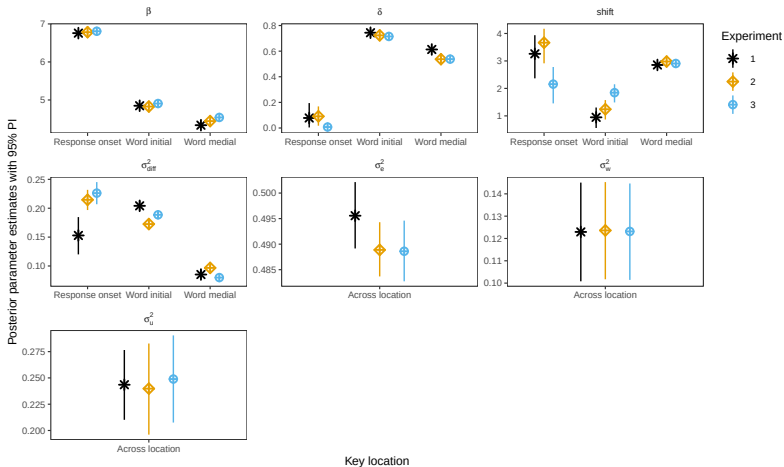
constraints:

$$\delta, \text{ (and all } \sigma\text{s)} > 0$$

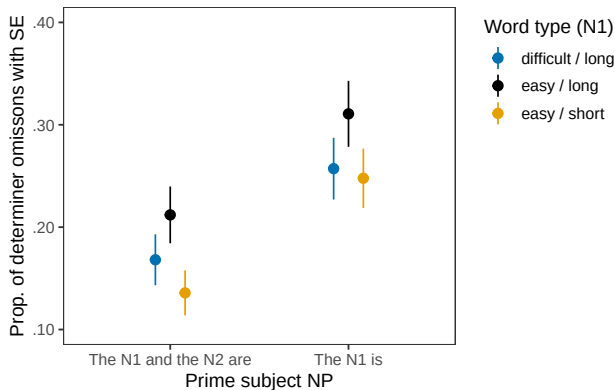
$$\sigma_{e'}^2 > \sigma_e^2$$

θ returns the proportion of hesitant inter-keystroke intervals for each key-location, prime and N1 condition.

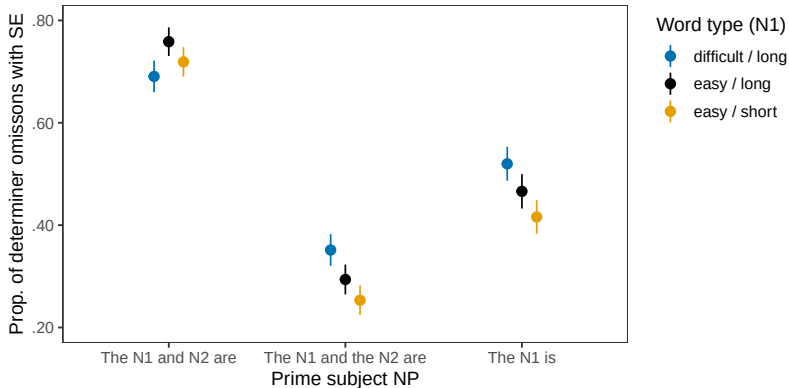
Most other mixture model parameter estimates



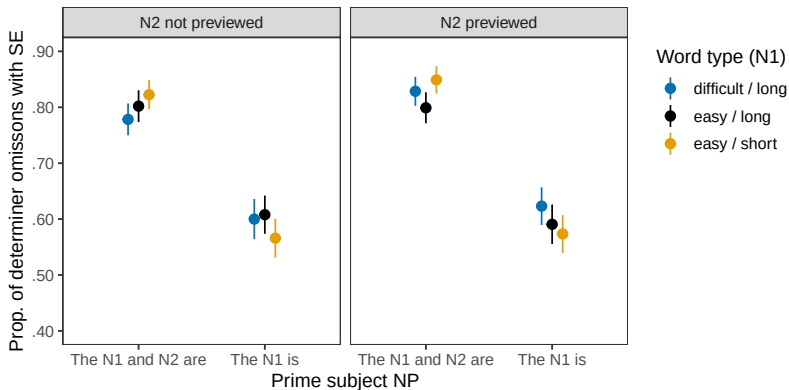
Experiment 1: determiner omissions before N2



Experiment 2: determiner omissions before N2



Experiment 3: determiner omissions before N2



Experiment 3: N2-initial hesitations

